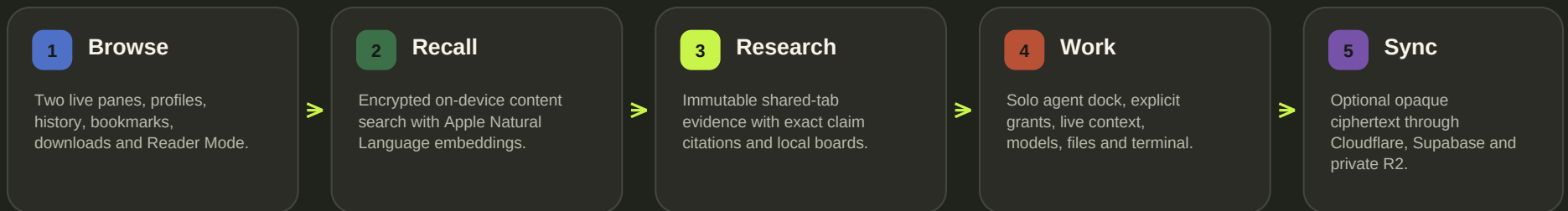




Locus Browser

Architecture, modularization, and performance guide - Intelligence and Productivity Canary



Browser first. Local intelligence by default. Explicit authority always.

This document describes the implemented Apple Silicon macOS canary and the boundaries that keep browsing, AI, and cloud services separate.

macOS 14+

Apple Silicon

Electron + React

Apache-2.0

Canary 0.1.0-canary.6



Processes, views, and authority boundaries

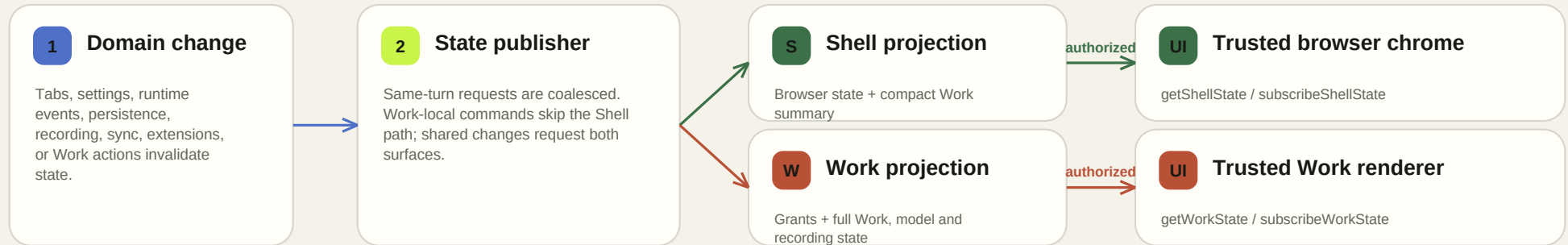




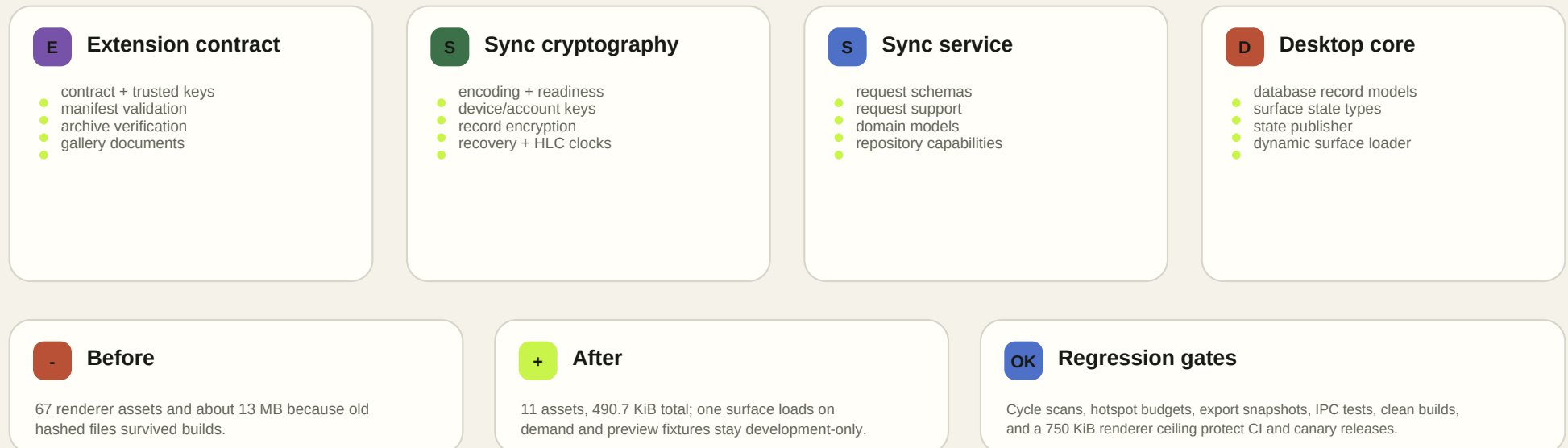
State paths, module seams, and build performance

Surface-specific state publication

A mutation is batched once, projected by trust boundary, and sent only to the renderer that consumes it.



Responsibility-focused modules behind stable entry points





Recall, Research Boards, and Tab Steward

R Private Semantic Recall

Opt-in on first use. Eligible pages visited after enablement are captured with strict extraction, canonical URL deduplication, encrypted text, full-text terms and one mean-pooled embedding.

- Searches open tabs, bookmarks and history with natural language and time hints.
- Apple Natural Language embeddings; deterministic keyword fallback for unsupported languages.
- 500 MB per-profile cap; oldest unbookmarked content is evicted first.
- Per-result delete, excluded sites, index size and Clear Recall Data controls.

C Cited Research Board

Up to ten explicitly shared current-window tabs become immutable encrypted snapshots. Passage ranking keeps total source text below 120,000 characters before a read-only model request.

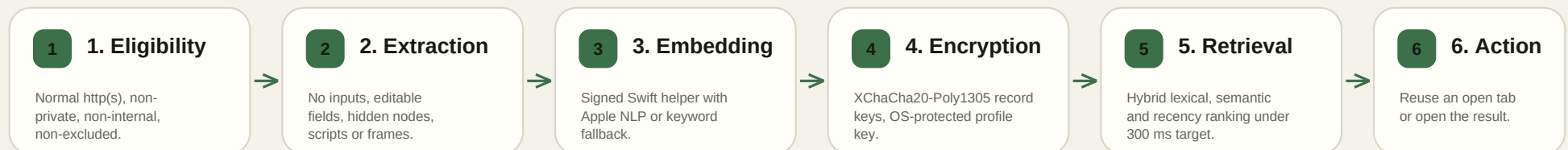
- Every factual claim must cite an exact source ID and passage ID.
- Uncited or invalid model output fails visibly instead of being displayed.
- Saved boards survive source navigation or closure and remain local to the profile.
- Exports sanitized Markdown footnotes or a rendered PDF through native save dialogs.

T AI Tab Steward

Provider-independent suggestions use canonical URLs, host and title similarity. Page text and tab metadata never leave the Mac.

- Quiet badges only for exact duplicates or high-confidence clusters of at least three.
- A full preview lists every move, group, rename and potential closure.
- Duplicate closure and Save-and-Close bundles each require separate confirmation.
- Resume Later reopens bundles without duplicating already-open canonical URLs.

Shared local intelligence pipeline





Split View, command palette, and Reader

2 Two-page Split View

Two live page panes share one window; the focused pane owns browser controls.

PRIMARY - focused

Selecting a tab replaces only the focused pane.

SECONDARY

Drag a tab onto L or R to assign it explicitly.

- Neither visible pane sleeps.
- Recording pauses before an unshared pane frame.
- Settings hides panes without destroying them.

K Universal Command Palette

Command-K searches trusted browser data and allowlisted actions in place.

Search tabs, history, settings, or run a command...

Google	open tab
Toggle Split View	command
Local models article	private recall

- Exact and fuzzy matches rank before semantics.
- Existing canonical URLs are reused instead of duplicated.
- Private windows expose only current private tabs and ephemeral actions.

A Reader Mode with Read Aloud

Safe extraction runs inside the isolated DOM bridge, then host-side sanitization applies a second strict allowlist. Reader replaces only the selected pane, so the other split page can remain live.

A quieter page for focused reading

Reader keeps the article central and trusted controls visible.

Read Aloud highlights the current sentence and uses voices already installed on this Mac.

- Locus, paper and dark appearances
- Adjustable text size, width and line spacing
- Links return through validated browser navigation
- Private windows keep article contents ephemeral
- Installed macOS system voices - no AI provider
- Rate, play/pause and previous/next sentence
- Current sentence highlighting
- Speech stops on navigation, crash, close or exit



What is stored, shared, synced, and excluded

LOCAL ENCRYPTED

Recall page text + vectors

Research boards + evidence

Resume Later bundles

Recording transcripts

Provider credentials

Sync account/device keys

NORMAL LOCAL

Tabs and groups

History and bookmarks

Downloads + permissions

Reader preferences

Extension installs

Crash restore state

OPTIONAL CLOUD

Bookmarks

History events

Tab groups

Remote-device tabs

Browser settings subset

Gallery extension metadata

NEVER CAPTURED

Passwords / passkeys

Payment fields

Credential inputs

Private windows

Internal / local files

Inaccessible frames

KEY OS-protected keys

Random per-profile content keys and recording keys are wrapped by macOS secure storage. XChaCha20-Poly1305 associated data binds every record to its profile, kind and ID.

DOM Strict extraction

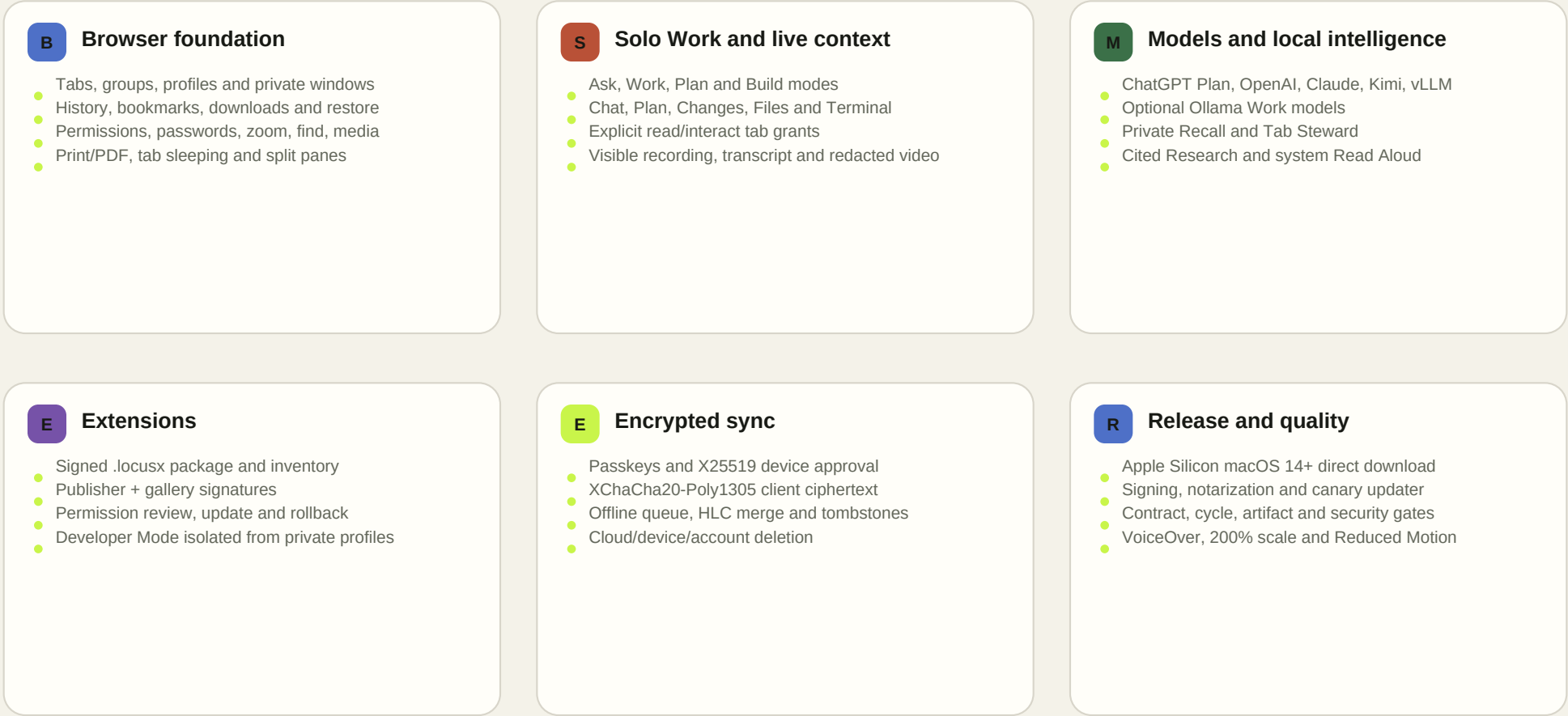
Forms, inputs, editable regions, credential/payment shapes, hidden content, scripts and inaccessible frames are excluded before text reaches Recall or Research.

STOP Pause instead of leak

Indexing pauses during recording, intensive agent work or serious thermal pressure. Recording pauses when a focused pane becomes unshared or protected.



Current feature inventory and service topology



Production service path

